

# Christopher A. Theissen

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| PROFESSIONAL<br>APPOINTMENTS   | Assistant Professor                                                                                            | Jul 2023–Present  |
|                                | UC San Diego, Department of Astronomy & Astrophysics                                                           |                   |
|                                | UCSD Chancellor’s Postdoctoral Fellow                                                                          | Sep 2021–Jun 2023 |
|                                | NASA Sagan Postdoctoral Fellow                                                                                 | Sep 2019–Sep 2022 |
|                                | Visting Scholar                                                                                                | Jan 2019–Sep 2019 |
|                                | UC San Diego, Center for Astrophysics and Space Sciences                                                       |                   |
|                                | Postdoctoral Scholar – Konopacky Group                                                                         | Jan 2018–Jan 2019 |
|                                | UC San Diego, Center for Astrophysics and Space Sciences<br>PI: Quinn Konopacky                                |                   |
| EDUCATION                      | Adjunct Professor – San Diego Mesa College                                                                     | Jun 2017–Jan 2019 |
|                                | Department of Physical Sciences                                                                                |                   |
|                                | <b>Boston University</b> , Boston, Massachusetts, USA                                                          |                   |
|                                | Doctor of Philosophy (Ph.D.) in Astronomy                                                                      | Jan 2018          |
|                                | Thesis: <i>Low-mass Stars with Extreme Mid-Infrared Excesses: Potential Signatures of Planetary Collisions</i> |                   |
|                                | Master of Arts (M.A.) in Astronomy                                                                             | May 2014          |
|                                | <b>University of California San Diego</b> , La Jolla, California, USA                                          |                   |
|                                | Bachelor of Science (B.S.) in Physics                                                                          | Jun 2010          |
| ACADEMIC<br>AWARDS &<br>HONORS | Specialization in Astrophysics                                                                                 |                   |
|                                | Bachelor of Arts (B.A.) in Mathematics                                                                         | Jun 2010          |
|                                | Applied Science                                                                                                |                   |
|                                | <b>San Diego Mesa College</b> , San Diego, California, USA                                                     |                   |
|                                | Associate of Arts (A.A.) in Transfer Studies                                                                   | Jun 2007          |
|                                | Hellman Fellow                                                                                                 | 2025–2026         |
|                                | Scialog Fellow, Early Science with the LSST                                                                    | 2024–2025         |
|                                | UC San Diego Chancellor’s Postdoctoral Fellowship                                                              | 2021–2023         |
| RESEARCH<br>EXPERIENCE         | UC San Diego Chancellor’s Outstanding Postdoctoral Scholar Award                                               | 2020–2021         |
|                                | NASA Hubble Fellowship Program Sagan Postdoctoral Fellowship                                                   | 2019–2022         |
|                                | National Science Foundation Graduate K–12 Fellowship                                                           | 2014–2015         |
|                                | Excellent Teaching Fellow Award, Boston University                                                             | 2012              |
|                                | Ford Foundation Predoctoral Fellowship                                                                         | 2012–2016         |
|                                | California Alliance for Minority Participation Graduate School Application Award                               | 2011              |
|                                | Minority Undergraduate Research Fellowship, California Institute of Technology                                 | 2009              |
|                                | Opportunity Grant, University of California San Diego                                                          | 2009–2010         |
| RESEARCH<br>EXPERIENCE         | <b>Konopacky Group</b> , UC SAN DIEGO                                                                          | 2018–2019         |
|                                | Postdoctoral Scholar                                                                                           |                   |
|                                | PI: Quinn Konopacky                                                                                            |                   |
|                                | <b>Cool Star Lab</b> , UC SAN DIEGO                                                                            | 2015–2017         |
|                                | Visiting Graduate Student Researcher                                                                           |                   |
|                                | Mentor: Adam Burgasser                                                                                         |                   |

TEACHING  
EXPERIENCE

|                                                                                                                   |             |
|-------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Astronomy 223: Astrostatistics</b> , UC SAN DIEGO, LA JOLLA, CA                                                | Winter 2026 |
| <b>Astronomy 150: Observational Optical Laboratory</b> , UC SAN DIEGO, LA JOLLA, CA                               | Winter 2025 |
| <b>Astronomy 20A: Introduction to Astrophysics I</b> , UC SAN DIEGO, LA JOLLA, CA                                 | Fall 2024   |
| <b>Physics 164: Observational Astrophysics Lab</b> , UC SAN DIEGO, LA JOLLA, CA                                   | Winter 2024 |
| <b>Physics 239: Astrophysical Statistics</b> , UC SAN DIEGO, LA JOLLA, CA                                         | Fall 2023   |
| <b>Astronomy 101: Descriptive Astronomy</b> , MESA COLLEGE, SAN DIEGO, CA<br>Adjunct Faculty                      | Summer 2017 |
| <b>8th Grade Science</b> , ATLANTIC MIDDLE SCHOOL, QUINCY, MASSACHUSETTS<br>Resident Scientist (NSF GK-12 Fellow) | 2014–2015   |
| <b>Astronomy 203: Principles of Astronomy II</b> , BOSTON UNIVERSITY<br>Teaching Fellow                           | Spring 2012 |
| <b>Astronomy 101: The Solar System</b> , BOSTON UNIVERSITY<br>Teaching Fellow                                     | Fall 2011   |

FIRST & SECOND  
AUTHOR  
REFEREED  
PUBLICATIONS

\*DIRECTLY MENTORED STUDENT CO-AUTHORS ARE UNDERLINED.

- A11 Zhou, T., **Theissen, C. A.** et al. (incl. Zhao, L.), “Classifying Cool Dwarfs: Comprehensive Spectral Typing of Field and Peculiar Dwarfs Using Machine Learning,” *Astrophysical Journal*, 992, 93, Oct 2025
- A10 Gan, T., **Theissen, C. A.** et al., “Metallicity Dependence of Giant Planets around M Dwarfs,” *Astrophysical Journal Supplement*, 276, 47, Feb 2025
- A9 Hoch, K. K. W., **Theissen, C. A.** et al., “JWST-TST High Contrast: Spectroscopic Characterization of the Benchmark Brown Dwarf HD 19467 B with the NIRSpec Integral Field Spectrograph,” *Astronomical Journal*, 168, 187, Oct 2024
- A8 Wei, L., **Theissen, C. A.** et al., “The 3D Kinematics of the Orion Nebula Cluster. II. Mass-dependent Kinematics of the Inner Cluster,” *Astrophysical Journal*, 962, 174, Feb 2024
- A7 **Theissen, C. A.** et al. (incl. Hsu, C., Wei, L.), “The 3-D Kinematics of the Orion Nebula Cluster: NIRSPA0 Radial Velocities of the Core Population,” *Astrophysical Journal*, 926, 141, Oct 2021
- A6 **Theissen, C. A.**, “Parallaxes of Cool Objects with *WISE*: Filling in for *Gaia*,” *Astrophysical Journal*, 862, 173, Aug 2018
- A5 **Theissen, C. A.** et al., “2MASS J11151597+1937266: A Young, Dusty, Isolated, Planetary-Mass Object with a Potential Wide Stellar Companion,” *Astrophysical Journal*, 853, 75, Jan 2018
- A4 **Theissen, C. A.** & West, A. A., “Collisions of Terrestrial Worlds: The Occurrence of Extreme Mid-Infrared Excesses around Low-mass Field Stars,” *Astronomical Journal*, 153, 165, Apr 2017
- A3 **Theissen, C. A.** et al. (incl. Shippee, G.), “The Late-Type Extension to MoVeRS (LaTE-MoVeRS): Proper Motion Verified Low-mass Stars and Brown Dwarfs from SDSS, 2MASS, and *WISE*,” *Astronomical Journal*, 153, 92, Feb 2017
- A2 **Theissen, C. A.**, West, A. A., & Dhital, S., “Motion Verified Red Stars (MoVeRS): A Catalog of Proper Motion Selected Low-mass Stars from *WISE*, SDSS, and 2MASS,” *Astronomical Journal*, 151, 41, Feb 2016
- A1 **Theissen, C. A.** & West, A. A., “Warm Dust around Cool Stars: *WISE* 12 and 22  $\mu\text{m}$  Excesses around SDSS M Dwarfs,” *Astrophysical Journal*, 794, 146, Oct 2014

CO-AUTHOR  
REFEREED  
PUBLICATIONS

- B61 Baburaj, A., et al. (incl. **Theissen, C. A.**), “A High-resolution Spectroscopic Survey of Directly Imaged Companion Hosts. II. Diversity in C/O Ratios among Host Stars,” *Astronomical Journal*, 171, 21, Jan 2026

- B60 Draxl Giannoni, J. D., et al. (incl. **Theissen, C. A.**), “Identifying and Characterizing Very Low Mass Spectral Blend Binaries with Machine Learning Methods,” *Astronomical Journal*, in press, Dec 2025
- B59 Millar-Blanchaer, M. A., et al. (incl. **Theissen, C. A.**), “The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. VI. Evidence for Radially Evolving Icy Grains in the HD 141569A Disk via NIRCcam Coronagraphic Imaging,” *Astrophysical Journal*, 994, 199, Nov 2025
- B58 Burgasser, A. J., et al. (incl. **Theissen, C. A.**), “Observation of undepleted phosphine in the atmosphere of a low-temperature brown dwarf,” *Science*, 390, 697, Nov 2025
- B57 Davoudi, F., et al. (incl. **Theissen, C. A.**), “Gravity-sensitive Spectral Indices in Ultracool Dwarfs: Investigating Correlations with Metallicity and Planet Occurrence Using SpeX and Fire Observations,” *Astronomical Journal*, 170, 213, Oct 2025
- B56 Gan, T., et al. (incl. **Theissen, C. A.**), “A New Brown Dwarf Orbiting an M Star and an Investigation of the Eccentricity Distribution of Transiting Long-period Brown Dwarfs,” *Astrophysical Journal Letters*, 988, 78, Aug 2025
- B55 Hoch, K. K. W., et al. (incl. **Theissen, C. A.**), “Silicate clouds and a circumplanetary disk in the YSES-1 exoplanet system,” *Nature*, 643, 938, Jul 2025
- B54 Zheng, W., et al. (incl. **Theissen, C. A.**), “SN 2023ixf in the Pinwheel Galaxy M101: From Shock Breakout to the Nebular Phase,” *Astrophysical Journal*, 988, 61, Jul 2025
- B53 Scott, M. G., et al. (incl. **Theissen, C. A.**), “TOI-6478 b: a cold underdense Neptune transiting a fully convective M dwarf from the thick disc,” *Monthly Notices of the Royal Astronomical Society*, 540, 1909, Jun 2025
- B52 Shrishmoy, R., et al. (incl. **Theissen, C. A.**), “Spectropolarimetric Evolution of SN 2023ixf: an Asymmetric Explosion in a Confined Aspherical Circumstellar Medium,” *Astrophysical Journal*, in press, Apr 2025
- B51 Shrishmoy, R., et al. (incl. **Theissen, C. A.**), “The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. III. Aperture Masking Interferometric Observations of the Star HIP 65426 at 3.8  $\mu\text{m}$ ,” *Astrophysical Journal Letters*, 983, 25, Apr 2025
- B50 Burgasser, A. J., et al. (incl. Karpoor, P. R., **Theissen, C. A.**), “New Cold Subdwarf Discoveries from Backyard Worlds and a Metallicity Classification System for T Subdwarfs,” *Astrophysical Journal Supplement*, 982, 79, Apr 2025
- B49 Barkaoui, K., et al. (**Theissen, C. A.**), “TOI-6508 b: A massive transiting brown dwarf orbiting a low-mass star,” *Astronomy & Astrophysics*, 696, 44, Apr 2025
- B48 Sappay, B. J., et al. (**Theissen, C. A.**), “HD 206893 B at High Spectral Resolution with the Keck Planet Imager and Characterizer,” *Astronomical Journal*, 169, 175, Mar 2025
- B47 Baburaj, A., et al. (incl. **Theissen, C. A.**), “A High-resolution Spectroscopic Survey of Directly Imaged Companion Hosts. I. Determination of Diagnostic Stellar Abundances for Planet Formation and Composition,” *Astronomical Journal*, 169, 55, Feb 2025
- B46 Hsu, C. C., et al. (incl. **Theissen, C. A.**), “The Brown Dwarf Kinematics Project (BDKP). VI. Ultracool Dwarf Radial and Rotational Velocities from SDSS/APOGEE High-resolution Spectroscopy,” *Astrophysical Journal Supplement*, 274, 40, Oct 2024
- B45 Franson, K., et al. (incl. **Theissen, C. A.**), “JWST/NIRCcam 4-5  $\mu\text{m}$  Imaging of the Giant Planet AF Lep b,” *Astrophysical Journal Letters*, 974, 11, Oct 2024
- B44 Burgasser, A. J., et al. (incl. **Theissen, C. A.**, Karpoor, P. R.), “Discovery of a Hypervelocity L Subdwarf at the Star/Brown Dwarf Mass Limit,” *Astrophysical Journal Letters*, 971, 25, Aug 2024
- B43 Ruffio, J. B., et al. (incl. **Theissen, C. A.**), “JWST-TST High Contrast: Achieving direct spectroscopy of faint substellar companions next to bright stars with the NIRSpec IFU,” *Astronomical Journal*, 168, 73, Aug 2024
- B42 Gillon, M., et al. (incl. **Theissen, C. A.**), “Detection of an Earth-sized exoplanet orbiting the nearby ultracool dwarf star SPECULOOS-3,” *Nature Astronomy*, 8, 865, Jul 2024

- B41 Barkaoui, K., et al. (incl. Karpoor, P. R., Softich, E., **Theissen, C. A.**), “Three short-period Earth-sized planets around M dwarfs discovered by TESS: TOI-5720b, TOI-6008b and TOI-6086b,” *Astronomy & Astrophysics*, 687, 264, Jul 2024
- B40 Do Ó, C. R., et al. (incl. **Theissen, C. A.**), “Orbital and Atmospheric Characterization of the 1RXS J034231.8+121622 System using High-resolution Spectroscopy Confirms that the Companion is a Low-mass Star,” *Astronomical Journal*, 167, 287, Jun 2024
- B39 Rothermich, A., et al. (incl. Softich, E., Karpoor, P. R., **Theissen, C. A.**), “89 New Ultracool Dwarf Comoving Companions Identified with the Backyard Worlds: Planet 9 Citizen Science Project,” *Astronomical Journal*, 167, 253, Jun 2024
- B38 Petrus, S., et al. (incl. **Theissen, C. A.**), “The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. V. Do Self-consistent Atmospheric Models Represent JWST Spectra? A Showcase with VHS 1256–1257 b,” *Astrophysical Journal Letters*, 966, 11, May 2024
- B37 Kirkpatrick, J. D., et al. (incl. **Theissen, C. A.**), “The Initial Mass Function Based on the Full-sky 20 pc Census of  $\sim 3600$  Stars and Brown Dwarfs,” *Astrophysical Journal Supplement*, 271, 55, Apr 2024
- B36 McCarthy, A. M., et al. (incl. **Theissen, C. A.**), “Multiple Patchy Cloud Layers in the Planetary-mass Object SIMP 0136+0933,” *Astrophysical Journal*, 965, 83, Apr 2024
- B35 Sallum, S., et al. (incl. **Theissen, C. A.**), “The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. IV. NIRISS Aperture Masking Interferometry Performance and Lessons Learned,” *Astrophysical Journal Letters*, 963, 2, Mar 2024
- B34 Ray, S., et al. (incl. **Theissen, C. A.**), “The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems III: Aperture Masking Interferometric Observations of the star HIP 65426 at  $3.8\ \mu\text{m}$ ,” *Astrophysical Journal Letters*, in press, Oct 2023
- B33 Vasylyev, S. S., et al. (incl. Karpoor, P. R., Softich, E. R., **Theissen, C. A.**), “Early Time Spectropolarimetry of the Aspherical Type II Supernova SN 2023ixf,” *Astrophysical Journal Letters*, 955, 37, Sep 2023
- B32 Jacobson-Galán, W. V., et al. (incl. Karpoor, P. R., Softich, E. R., **Theissen, C. A.**), “SN 2023ixf in Messier 101: Photo-ionization of Dense, Close-in Circumstellar Material in a Nearby Type II Supernova,” *Astrophysical Journal Letters*, 954, 42, Sep 2023
- B31 Hoch, K. K. W., et al. (incl. **Theissen, C. A.**), “Assessing the C/O Ratio Formation Diagnostic: A Potential Trend with Companion Mass,” *Astronomical Journal*, 166, 85, Sep 2023
- B30 Ghachoui, M., et al. (incl. Aganze, C., Gerasimov, R., Hsu, C., **Theissen, C. A.**), “TESS discovery of a super-Earth orbiting the M-dwarf star TOI-1680,” *Astronomy & Astrophysics*, 677, A31, Sep 2023
- B29 Franson, K., et al. (incl. **Theissen, C. A.**), “Astrometric Accelerations as Dynamical Beacons: A Giant Planet Imaged Inside the Debris Disk of the Young Star AF Lep,” *Astrophysical Journal Letters*, 950, 19, Jun 2023
- B28 Pozuelos, F. J., et al. (incl. Aganze, C., Gerasimov, R., **Theissen, C. A.**), “A super-Earth and a mini-Neptune near the 2:1 MMR straddling the radius valley around the nearby mid-M dwarf TOI-2096,” *Astronomy & Astrophysics*, 672, A70, Apr 2023
- B27 Hsu, C., Burgasser, A. J., & **Theissen, C. A.**, “Discovery of the Exceptionally Short Period Ultracool Dwarf Binary LP 413-53AB,” *Astrophysical Journal Letters*, 945, 6, Mar 2023
- B26 Schneider, A. C., et al. (incl. Aganze, C., **Theissen, C. A.**), “Redder than Red: Discovery of an Exceptionally Red L/T Transition Dwarf,” *Astrophysical Journal Letters*, 943, 16, Feb 2023
- B25 Franson, K., et al. (incl. **Theissen, C. A.**), “Astrometric Accelerations as Dynamical Beacons: Discovery and Characterization of HIP 21152 B, the First T-Dwarf Companion in the Hyades,” *Astronomical Journal*, 165, 39, Feb 2023
- B24 Tamburo, P., et al. (incl. **Theissen, C. A.**), “The Perkins INfrared Exosatellite Survey (PINES) II. Transit Candidates and Implications for Planet Occurrence around L and T Dwarfs,” *Astronomical Journal*, 164, 252, Dec 2022

- B23 Delrez, L., et al. (incl. Aganze, C., **Theissen, C. A.**), “Two Temperate Super-Earths Transiting a Nearby Late-type M Dwarf,” *Astronomy & Astrophysics*, 667, 59, Nov 2022
- B22 Hoch, K. W. et al. (incl. **Theissen, C. A.**), “Moderate-Resolution *K*-Band Spectroscopy of the Substellar Companion VHS 1256 b,” *Astronomical Journal*, 164, 155, Oct 2022
- B21 Gan, T., et al. (incl. **Theissen, C. A.**, Aganze, C. A.), “TESS discovery of a sub-Neptune orbiting a mid-M dwarf TOI-2136,” *Monthly Notices of the Royal Astronomical Society*, 514, 4120, Aug 2022
- B20 Aganze, C. et al. (incl. **Theissen, C. A.**, Tejada Arevalo, R. A., Hsu, C.), “Beyond the Local Volume II: Population Scaleheights and Ages of Ultracool Dwarfs in Deep HST/WFC3 Parallel Fields,” *Astrophysical Journal*, 934, 73, Jul 2022
- B19 Tamburo, P., et al. (incl. **Theissen, C. A.**), “The Perkins INfrared Exosatellite Survey (PINES) I. Survey Overview, Reduction Pipeline, and Early Results,” *Astronomical Journal*, 163, 253, Jun 2022
- B18 Abdurro’uf et al. (incl. **Theissen, C. A.**), “The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar and APOGEE-2 Data,” *Astrophysical Journal Supplement Series*, 259, 35, Apr 2022
- B17 Aganze, C. et al. (incl. **Theissen, C. A.**, Tejada Arevalo, R. A., Hsu, C.), “Beyond the Local Volume: Surface Densities of Ultracool Dwarfs in Deep HST/WFC3 Parallel Fields,” *Astrophysical Journal*, 924, 114 Jan 2022
- B16 Hsu, C. et al. (incl. **Theissen, C. A.**, Birky, J., Aganze, C.), “The Brown Dwarf Kinematics Project (BDKP). V. Radial and Rotational Velocities of T Dwarfs from Keck/NIRSPEC High-Resolution Spectroscopy,” *Astrophysical Journal*, 257, 45, Dec 2021
- B15 Faherty, J. K., et al. (incl. Aganze, C., Hsu, C., Gerasimov, R., **Theissen, C. A.**), “A Wide Planetary Mass Companion Discovered Through the Citizen Science Project Backyard Worlds: Planet 9,” *Astrophysical Journal*, 923, 48, Dec 2021
- B14 Schneider, A. C., et al. (incl. Aganze, C., Gerasimov, R., Hsu, C., **Theissen, C. A.**), “Ross 19B: An Extremely Cold Companion Discovered via the Backyard Worlds: Planet 9 Citizen Science Project,” *Astrophysical Journal*, 920, 140, Nov 2021
- B13 Wells, R. D., et al. (incl. **Theissen, C. A.**), “A large sub-Neptune transiting the thick-disk M4V TOI-2406,” *Astronomy & Astrophysics*, 653, A97 Sep 2021
- B12 Meisner, A. M., et al. (incl. **Theissen, C. A.**, Gerasimov, R., Aganze, C., Hsu, C.), “New Candidate Extreme T Subdwarfs from the Backyard Worlds: Planet 9 Citizen Science Project,” *Astrophysical Journal*, 915, 120, May 2021
- B11 Kirkpatrick, J. D., et al. (incl. Aganze, C., Gerasimov, R., Hsu, C., **Theissen, C. A.**), “The Field Substellar Mass Function Based on the Full-sky 20-pc Census of 525 L, T, and Y Dwarfs,” *Astrophysical Journal Supplement Series*, 253, 7, May 2021
- B10 Wilcomb, K. K. et al. (incl. **Theissen, C. A.**), “Moderate-Resolution *K*-Band Spectroscopy of Substellar Companion  $\kappa$  Andromedae b,” *Astronomical Journal*, 160, 270, Nov 2020
- B9 Meisner, A. M., et al. (incl. Hsu, C., Aganze, C., Gerasimov, R., **Theissen, C. A.**), “*Spitzer* Follow-up of the Coldest Brown Dwarfs Discovered by the Backyard Worlds: Planet 9 Citizen Science Project,” *Astrophysical Journal*, 889, 123, Aug 2020
- B8 Muirhead, P. S., et al. (incl. **Theissen, C. A.**), “Magnetic Inflation and Stellar Mass. V. Intensification and saturation of M dwarf absorption lines with Rossby number,” *Astronomical Journal*, 159, 52, Feb 2020
- B7 Bardalez Gagliuffi, D. C., et al. (incl. **Theissen, C. A.**), “The Ultracool SpeXtroscopic Survey. I. Volume-limited Spectroscopic Sample and Luminosity Function of M7–L5 Ultracool Dwarfs,” *Astrophysical Journal*, 883, 205, Oct 2019
- B6 Kim, D., et al. (incl. **Theissen, C. A.**), “Stellar Proper Motions in the Orion Nebular Cluster,” *Astronomical Journal*, 157, 118, Feb 2019

- B5 Gagné, J., et al. (incl. **Theissen, C. A.**), “2MASS J13243553+6358281 is an Early T-Type Planetary-mass Object in the AB Doradus Moving Group,” *Astrophysical Journal Letters*, 854, 27, Feb 2018
- B4 Favia, A., West, A. A., & **Theissen, C. A.**, “Runaway M Dwarf Candidates from the Sloan Digital Sky Survey,” *Astrophysical Journal*, 813, 26, Nov 2015
- B3 Arcavi, I., et al. (incl. **Theissen, C. A.**), “A Continuum of H- to He-rich Tidal Disruption Candidates With a Preference for E+A Galaxies,” *Astrophysical Journal*, 793, 38, Sep 2014
- B2 Sfiligoi, I., et al. (incl. **Theissen, C. A.**), “Scalability of network facing services used in the Open Science Grid,” *Journal of Physics: Conference Series*, 331, 062023, Dec 2011
- B1 Quimby, R. M., et al. (incl. **Theissen, C. A.**), “Hydrogen-poor superluminous stellar explosions,” *Nature*, 474, 487, Jun 2011

**PREPRINTS &  
OTHERS**

- C16 Loritsch, M., et al. (incl. **Theissen, C. A.**, Karpoor, P.), “Kast Optical Spectra of Unclassified Stars Within 20 pc of the Sun,” *Research Notes of the American Astronomical Society*, 9, 125, May 2025.
- C15 Alvarado, E., et al. (incl. **Theissen, C. A.**), “The Spectral ANalog of Dwarfs (SAND): New Model Atmospheres with Varying Chemistry for Galactic Archaeology with Ultracool Dwarfs,” *Research Notes of the American Astronomical Society*, 8, 134, May 2024.
- C14 Zhou, T., et al. (incl. **Theissen, C. A.**), “Spectral Typing with Artificial Intelligence: Classifying Low-resolution Near-infrared Spectra of Standard M/L/T Dwarfs,” *Research Notes of the American Astronomical Society*, 8, 102, Apr 2024.
- C13 Mainieri, V., et al. (incl. **Theissen, C. A.**), “The Wide-field Spectroscopic Telescope (WST) Science White Paper,” *White Paper*, Mar 2024.
- C12 Humphreys, A., et al. (incl. **Theissen, C. A.**), “Spectroscopic Confirmation of the Nearby, Wide-separation L Dwarf Pair CWISE J061741.79+194512.8AB,” *Research Notes of the American Astronomical Society*, 7, 184, Jul 2023.
- C11 Dage, K. C., et al. (incl. **Theissen, C. A.**), “Extragalactic Star Cluster Science with the Nancy Grace Roman Space Telescope’s High Latitude Wide Area Survey and the Vera C. Rubin Observatory,” *White Paper for the Roman Core Community Survey, submitted to BAAS*, Jun 2023.
- C10 Han, J. J., et al. (incl. **Theissen, C. A.**), “NANCY: Next-generation All-sky Near-infrared Community survey,” *White Paper for the Roman Core Community Survey, submitted to BAAS*, Jun 2023.
- C9 Zhou, T., et al. (incl. **Theissen, C. A.**), “Resolved Binaries with Late-M and L Dwarf Companions Identified in Gaia eDR3,” *Research Notes of the American Astronomical Society*, 7, 50, Mar 2023.
- C8 Desai, M., et al. (incl. Draxl Giannoni, J. D., Aganze, C., **Theissen, C. A.**), “Identifying Ultracool Binary Systems using Machine Learning Methods,” *Research Notes of the American Astronomical Society*, 6, 151, Jan 2023.
- C7 **Theissen, C. A.**, et al. (incl. Hsu, C.), “Keck NIREs Spectral Standards for L, T, & Y Dwarfs,” *Research Notes of the American Astronomical Society*, 6, 151, Jul 2022.
- C6 Low, R., et al. (incl. Gerasimov, R., Hsu, C., **Theissen, C. A.**), “Spectroscopic Confirmation of an M6 Dwarf Companion to the Nearby Star BD-08 2582,” *Research Notes of the American Astronomical Society*, 5, 26, Feb 2021.
- C5 **Theissen, C. A.**, et al., “WISE J135501.90-825838.9 is a Nearby, Young, Extremely Low-mass Substellar Binary,” *Research Notes of the American Astronomical Society*, 4, 67, May 2020.
- C4 Muirhead, P. S., et al. (incl. **Theissen, C. A.**), “Searching for Exosatellites Orbiting L and T Dwarfs: Connecting Planet Formation to Moon Formation and Finding New Temperate Worlds,” *Bulletin of the American Astronomical Society*, Astro2020 White Paper, 2019.
- C3 Dupuy, T. J., et al. (incl. **Theissen, C. A.**), “Establishing an Empirical Substellar Sequence to Planetary Masses,” *Bulletin of the American Astronomical Society*, Astro2020 White Paper, 2019.

- C2 Kirkpatrick, J. D., et al. (incl. **Theissen, C. A.**), “The Need for Infrared Astrometry of Brown Dwarfs in the Post-Gaia Era,” *Bulletin of the American Astronomical Society*, Astro2020 White Paper, 2019.
- C1 Burgasser, A. J., **Theissen, C. A.**, et al., “Identification of WISE J000100.45+065259.6 as an M8.5+T5 Spectral Binary Candidate,” *Research Notes of the American Astronomical Society*, 1, 47, Dec 2017.

#### CONFERENCE PROCEEDINGS

- D8 Hoch, K., et al. (incl. **Theissen, C. A.**), “Direct Imaging Spectroscopy of Substellar Companions with JWST,” *Extreme Solar Systems V, Bulletin of the American Astronomical Society*, 2024.
- D7 McCarthy, A., et al. (incl. **Theissen, C. A.**), “Multiple Patchy Cloud Layers in the Planetary Mass Object SIMP0136+0933,” *Extreme Solar Systems V, Bulletin of the American Astronomical Society*, 2024.
- D6 Baburaj, A., et al. (incl. **Theissen, C. A.**), “Constraining Formation of Directly Imaged Planets through High-Resolution Spectroscopy of Host Stars,” *Extreme Solar Systems V, Bulletin of the American Astronomical Society*, 2024.
- D5 Franson, K., et al. (incl. **Theissen, C. A.**), “Astrometric Accelerations as Dynamical Beacons: Efficiently Imaging Planets Around Young Accelerating Stars,” *Extreme Solar Systems V, Bulletin of the American Astronomical Society*, 2024.
- D4 Wilcomb, K. K., et al. (incl. **Theissen, C. A.**), “Moderate Resolution Spectroscopy of Directly Imaged Planets,” *Extreme Solar Systems IV, Bulletin of the American Astronomical Society*, 2019.
- D3 Burgasser, A. J. & **SPLAT Development Team**, “The SpeX Prism Library Analysis Toolkit (SPLAT): A Data Curation Model,” *3rd International Workshop on Spectral Stellar Libraries*, 14, 7-22, Oct 2017.
- D2 Chakrabarti, S., et al. (incl. **Theissen, C. A.**), “Laboratory and Field tests of a High Throughput and Multi-slit Imaging Spectrograph (HiT&MIS),” *39th COSPAR Scientific Assembly*, 293, Feb 2016.
- D1 Sfiligoi, I., Würthwein, F., & **Theissen, C. A.**, “Using Condor Glideins for Distributed Testing of Network Facing Services,” *Third International Joint Conference on Computational Science and Optimization*, 327-331, May 2010.

#### PRESENTATIONS & POSTERS

- “Are the Lowest Mass Planet Hosts Odd?: Open Questions Surrounding the Smallest Planet Hosts in Our Galaxy,”  
*University of Florida Astronomy Colloquium* [**Invited Talk**], Aug 2025.
- “Using the Smallest Stars to Explore Large-scale Habitability within the Milky Way Galaxy,”  
*Notre Dame Astrophysics Seminar* [**Invited Talk**], Nov 2022.  
*Carnegie EPL Seminar* [**Invited Talk**], Feb 2022.  
*UCSD Scripps IGPP Seminar* [**Invited Talk**], Feb 2022.  
*UCSD Physics Colloquium* [**Invited Talk**], Jan 2021.
- “The Three Dimensional Kinematics of the Low-mass Population within the ONC Core,”  
*NASA Hubble Fellowship Program Symposium* [Talk], Nov 2021.
- “Investigating Spectral Peculiarities in the Lowest-mass Planet Hosts,”  
*AAS 237 (Winter Meeting)* [Talk], Jan 2021.
- “Spectral Peculiarities in Ultracool Dwarf Planet Hosts,”  
*NASA SOFIA Colloquium* [**Invited Talk**], Nov 2020.
- “The Dos and Don’ts of Writing a Successful (Fellowship) Application,”  
*UCSD IDEA Center Postdoc Talk* [**Invited Talk**], Oct 2020.
- “The Connection Between Ultracool Dwarf Planet Hosts and Surface Gravity,”  
*NASA Hubble Fellowship Program Symposium* [Talk], Sep 2020.
- “3-D Kinematics in the ONC Core,”  
*Keck Science Meeting* [Poster], Sep 2020.

- “Supporting BIPOC Scientists through NHFP-organized Mentorship and Outreach,”  
*NASA Hubble Fellowship Program Symposium* [Talk], Sep 2020.
- “Multiplicity at the Bottom of the Main Sequence,”  
*San Diego State University Astronomy & Physics Colloquium* [**Invited Talk**], Nov 2019.
- “Planetary Collisions around Low-mass Stars: Constraining the Timescale for Collisions and Testing the Origin of the *Kepler* Dichotomy,”  
*NASA Hubble Fellowship Program Symposium* [Talk], Oct 2019.
- “Cooler than *Gaia*: Parallaxes of Ultracool Objects with *WISE*,”  
*UC San Diego Astrophysics Seminar* [**Invited Talk**], May 2018.
- “Low-mass Stars with Extreme Mid-Infrared Excesses: Potential Signatures of Planetary Collisions,”  
*AAS 231 (Winter Meeting)* [Talk], Jan 2018.
- “Exoplanets and the Search for Life around Low-mass Stars,”  
*Mesa College STEM Lecture Series* [**Invited Talk**], Nov 2017.
- “Cool Stars with Extreme Mid-Infrared Excesses: Potential Tracers of Planetary Collisions,”  
*AAS 228 (Summer Meeting) + Cool Stars 19* [Poster], 2016.
- “The Motion Verified Red Stars (MoVeRS) Catalog and Low-Mass Field Stars with Warm Dust,”  
*AAS 227 (Winter Meeting)* [Poster], 2016.
- “The Occurrence of Warm Dust around Cool Stars,”  
*UC San Diego CASS Journal Club* [Talk], 2015.
- “*WISE* Infrared Excess Detections for SDSS M Dwarfs: Cool Field Stars with Evidence of Warm Circumstellar Material,”  
*AAS 224 (Summer Meeting) + Cool Stars 18* [Poster], 2014.
- “SDSS M dwarfs with *WISE* Signatures of Infrared Excess: Evidence of Warm Circumstellar Material in Low-Mass Field Populations,”  
*AAS 223 (Winter Meeting)* [Poster], 2014.
- “GlideTester - A framework for distributed testing of network-facing services using Condor glideins on Grid resources,”  
*TeraGrid Conference* [Poster], 2010.
- “PyTracker: Automated Spectroscopic Target Acquisition using Cross-Correlation with Existing Astrometric Positions,”  
*University of California San Diego Undergraduate Research Conference* [Talk], 2010.
- “Automated Cross-Correlative Spectroscopic Analysis of the Optical Transient Sky via Images Acquired using the Palomar Transient Factory,”  
*California Institute of Technology Summer Seminar* [Poster], 2009.

## PRESS COVERAGE

### Ultrashort Period Ultracool Binary

- “Ultracool Dwarf Binary Stars Break Records”  
*Keck Observatory*, Feb 2023.
- “Record breakers! Super-close dwarf stars orbit each other in less than a day”  
*Space.com*, Feb 2023.
- “Astronomers spot a tiny binary”  
*Sky & Telescope*, Jan 2023.
- “Astronomers Discover Two Invisible Stars Spinning Around Each Other at Breakneck Speed”  
*Gizmodo*, Jan 2023.
- “Binary Dwarf Stars Found Orbiting Each Other Every 20 Hrs. They Were Once Almost Touching”  
*Universe Today*, Jan 2023.

### Extremely Low-mass Binary

- “*WISE* J135501.90-825838.9 is a young, extremely low-mass substellar binary, study finds”  
*Phys.org*, Mar 2020.

### Planetary Collisions

- “Some planets ripe for life may be doomed by billions of years of violent collisions”  
*Astronomy Magazine*, Jul 2016.



|                                   |                                                                                                                                                                                       |                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>RESEARCH<br/>COLLABS</b>       | Institutional Representative for UCSD on the LSST-Discovery Alliance Board                                                                                                            |                   |
|                                   | Member of the LSST Stars, Milky Way & Local Volume Science Collaboration                                                                                                              |                   |
|                                   | Member of the TESS Follow-up Observing Program (TFOP) Working Group (WG)                                                                                                              |                   |
|                                   | Member of the Wide-Field Spectroscopic Telescope Science Team                                                                                                                         |                   |
|                                   | Member of the Perkins INfrared Exosatellite Survey (PINES)                                                                                                                            |                   |
| <b>SERVICE &amp;<br/>OUTREACH</b> | <b>Keck Galactic TAC - Member,</b><br>UCO                                                                                                                                             | Feb 2025–Present  |
|                                   | <b>LSST-Discovery Alliance Catalyst Steering Committee - Member,</b><br>LSST-DA                                                                                                       | Nov 2024–Present  |
|                                   | <b>Cal-Bridge - Mentor,</b><br>UC SAN DIEGO                                                                                                                                           | Sep 2023–Present  |
|                                   | <b>Cosmic Tours - Volunteer/Presenter,</b><br>UC SAN DIEGO                                                                                                                            | Oct 2022–Present  |
|                                   | <b>Research Affairs Advisory Committee - Member,</b><br>UC SAN DIEGO                                                                                                                  | Oct 2022–Present  |
|                                   | <b>Robotics Olympiad - Coach,</b><br>SOLANA HIGHLANDS ELEMENTARY SCHOOL                                                                                                               | Nov 2025–Apr 2026 |
|                                   | <b>Science Olympiad - Coach,</b><br>SOLANA HIGHLANDS ELEMENTARY SCHOOL                                                                                                                | Oct 2025–Feb 2026 |
|                                   | <b>Yo Soy STEM Plática - Speaker,</b><br>“Navigating Community College (Or: Figuring out the rest of your life)”,<br>SAN DIEGO CITY COLLEGE                                           | Aug 2025          |
|                                   | <b>Cool Stars 22 Meeting - SOC Chair and LOC Chair,</b><br>UC SAN DIEGO                                                                                                               | Jun 2022–Jun 2024 |
|                                   | <b>NASA Hubble Fellowship Program Anti-Racism Initiative - Co-organizer,</b><br>LEAD FOR THE MENTORING AND OUTREACH SUBGROUP                                                          | 2020–2022         |
|                                   | <b>NHFP DEI Session - Presenter,</b><br>NASA HUBBLE FELLOWSHIP PROGRAM SYMPOSIUM                                                                                                      | Nov 2021          |
|                                   | <b>2021 Chancellor’s Award Review Committee - Reviewer,</b><br>UC SAN DIEGO                                                                                                           | Aug 2021          |
|                                   | <b>Coalition NSF Advocacy Day - Advocate,</b><br>REMOTE MEETINGS                                                                                                                      | Jun 2021          |
|                                   | <b>UCSD Spring STEM Transfer Seminar 2021 - Panelist,</b><br>“Traversing Postdocs and Industry Positions”,<br>UC SAN DIEGO                                                            | May 2021          |
|                                   | <b>UCSD Career Center - Panelist,</b><br>“PhD Career Summit: Applying to Postdoc Positions”,<br>UC SAN DIEGO                                                                          | Mar 2021          |
|                                   | <b>UCSD IDEA Center - Postdoctoral Scholarly Talks,</b><br>“Developing a Successful Ford Fellowship Application”,<br>UC SAN DIEGO                                                     | Oct 2020          |
|                                   | <b>Summer Training Academy for Research Success (STARS)/</b><br><b>California-Arizona Minority Partnership for Astronomy</b><br><b>Research and Education (CAMPARE), UC SAN DIEGO</b> | 2018, 2019, 2020  |
|                                   | <b>High Tech High Internship, UC SAN DIEGO</b>                                                                                                                                        | Jan 2020          |
|                                   | <b>Career Paths Session - Chair/Panelist,</b><br>NASA HUBBLE FELLOWSHIP PROGRAM SYMPOSIUM                                                                                             | Oct 2019          |

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|------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Institute for Scientist & Engineer Educators (ISEE)<br>Professional Development Program (PDP),<br>UC SANTA CRUZ/UC SAN DIEGO | 2018              |
| InterTribal Youth/Young Native Scholars Summer Program,<br>UC SAN DIEGO                                                      | Jul 2016          |
| Cal-Bridge Workshop on Graduate School, UC SAN DIEGO                                                                         | May 2016          |
| STEM Fest, VISTA HIGH SCHOOL                                                                                                 | Mar 2016          |
| High School Science Olympiad Coach, UNIVERSITY HIGH SCHOOL                                                                   | Oct 2015–Feb 2016 |
| Chambliss Award Judge, AMERICAN ASTRONOMICAL SOCIETY MEETING 227                                                             | Jan 2016          |
| Program on Student Success in Engineering (POSSE),<br>UNIVERSITY OF CALIFORNIA SAN DIEGO/GOMPERS PREPARATORY ACADEMY         | Sep 2015–Jun 2016 |
| Upward Bound, BOSTON UNIVERSITY                                                                                              | 2013–2015         |
| Research in Science and Engineering (RISE), BOSTON UNIVERSITY                                                                | 2013–2015         |
| U-Design, BOSTON UNIVERSITY, DEPARTMENT OF ENGINEERING                                                                       | Jul 2014          |
| Academy of the Pacific Rim Astronomy Day, BOSTON UNIVERSITY                                                                  | Nov 2012          |
| Graduate Women in Science and Engineering (GWISE) -<br>“How to Find a Fellowship” Panelist, BOSTON UNIVERSITY                | Sep 2012          |
| <i>Other:</i>                                                                                                                |                   |
| • OPTICON-Radionet Pilot TAC                                                                                                 | 2023              |
| • Subject-matter expert reviewer in a NASA peer review                                                                       | 2020, 2021, 2022  |
| • Panelist for a NASA review                                                                                                 | 2023              |
| • Panelist for a NASA review                                                                                                 | 2021              |
| Referee: The Astrophysical Journal (ApJ)                                                                                     | 2018–Present      |
| The Astrophysical Journal Supplement Series (ApJS)                                                                           |                   |
| Monthly Notices of the Royal Astronomical Society (MNRAS)                                                                    |                   |

## MENTORSHIP

|                                                              |              |
|--------------------------------------------------------------|--------------|
| <b>POSTDOCS</b>                                              |              |
| Rayna Rampalli, UCSD                                         | 2026–Present |
| Lyra Cao, UCSD                                               | 2026–Present |
| Julie Inglis, UCSD                                           | 2025–Present |
| <b>GRADUATES</b>                                             |              |
| Ana Victoria Ojeda Vera, CITEDIPN (Visiting Masters Student) | 2026–Present |
| Arthur Berberyan, UCSD                                       | 2025–Present |
| Phil Van-Lane, U. Toronto (Visiting Graduate Student)        | 2025–Present |
| Sarah Jiang, UCSD                                            | 2024–Present |
| Aman Parikh, UCSD (Masters in Data Science)                  | 2023–2024    |
| Preethi Karpoor, UCSD (Masters in Astronomy)                 | 2022–2024    |
| Aneesh Baburaj, UCSD (→ Postdoc @ Northwestern)              | 2021–2025    |
| Lingfeng Wei, UCSD (→ Postdoc @ UCLA)                        | 2020–2025    |
| Christian Aganze, UCSD (→ Science Fellow @ Stanford)         | 2018–2023    |
| Kielan Wilcomb, UCSD (→ Giacconi Fellow @ STScI)             | 2018–2022    |
| Chih-Chun Hsu, UCSD (→ Postdoc @ Northwestern)               | 2018–2022    |
| <b>UNDERGRADUATES</b>                                        |              |
| Madison Fierro, UCSD                                         | 2026–Present |
| Madhav Krishna, UCSD                                         | 2026–Present |
| Ily Rubio, UWYO (STARS)                                      | 2026–Present |
| Brighton Liu, UCSD                                           | 2026–Present |
| Johanna Sanchez, UCSD                                        | 2026–Present |

|                                                                |              |
|----------------------------------------------------------------|--------------|
| Tue Thai, UCSD                                                 | 2025–Present |
| Julia Zhang, UCSD                                              | 2025–Present |
| Barron Nguyen, Stanford (→ NSF @ Stanford)                     | 2025–2026    |
| Piper Lentz, Amherst College (STARS → Northeastern)            | 2025–2026    |
| Nishtha Panging, UCSD                                          | 2025–2026    |
| Aadhithiya Anbalagan, UCSD (STARTastro)                        | 2025         |
| Aditya Rao, SJSU (Cal-Bridge Scholar)                          | 2025         |
| Danny Nguyen, UMass Amherst                                    | 2025         |
| Tianxing “Sky” Zhou, UCSD (→ BYU)                              | 2022–2025    |
| Hanan Sayes, UCSD                                              | 2024–2025    |
| Jackie Scullin, UCSD                                           | 2023–2025    |
| Ashleann Chen, UCSD                                            | 2024         |
| Jacob Craighead, UCSD                                          | 2024         |
| Weston Chester, UCSD                                           | 2024         |
| So Hirota, UCSD (→ Stanford)                                   | 2023–2024    |
| Lexu “Gavin” Zhao, UCSD (→ UFL)                                | 2022–2024    |
| Malina Desai, UCSD (→ Carnegie post-bac → MIT)                 | 2022–2023    |
| Chelsea Adelman, Cal Poly Pomona (Cal-Bridge Scholar → UCI)    | 2020         |
| Roberto Tejada Arevalo, CSULA (Cal-Bridge Scholar → Princeton) | 2018–2020    |
| Dennis H. Calderon, CSUEB (Cal-Bridge Scholar → OSU)           | 2018–2019    |
| Russell Van Linge, UCSD (→ Scripps Health)                     | 2018         |
| Jessica Birky, UCSD (→ NSF @ UW)                               | 2016–2019    |
| Guillaume Shippee, UCB (→ Qualcomm)                            | 2016         |
| <b>HIGH SCHOOL STUDENTS</b>                                    |              |
| Siddharth Thambidurai, Intern                                  | 2026         |
| Sven Andersen, Intern (→ UCSD)                                 | 2022         |
| Angeli Solis, Intern (→ UCSD)                                  | 2020         |
| Victor Zhang, BU RISE (Princeton → U. Chicago)                 | 2015         |
| Katie Melbourne, BU RISE (Yale → Ball Aerospace)               | 2014         |
| Isabella Trierweiler, BU RISE (Yale → UCLA)                    | 2013         |

**GRANTS,  
FUNDING, &  
COMPUTE**

|                                                                                                                                                                                                      |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>The Better Benchmark Bureau: A New Generation of Wide Binary Systems to Characterize the Most Abundant Planet Hosts in the Galaxy</b><br>PI, Gordon and Betty Moore Foundation Fellowship, \$171k | 2026–2028 |
| <b>Finding Stellar Dance Partners: Identifying the Rarest Multiple Systems as Optimal Benchmarks for the Smallest Stars</b><br>PI, Hellman Fellowship, \$50k                                         | 2025–2027 |
| <b>Developing Simulations and Real-Time Alert Systems for Rare Transiting Events: Exoplanets around Brown Dwarfs as a Case Study</b><br>PI, LSST-DA Science Catalyst Small Grant, \$6k               | 2024–2025 |
| <b>It Takes Two Planets to Tango: Constraining the Orbit of a Planetary-Mass Binary</b><br>PI, HST Cycle 32 (& 33) Guest Observer, \$100k                                                            | 2024–2026 |
| <b>A Magnitude Limited Sample of M dwarfs to Study the Super-Earth Rate across the Fully Convective Boundary</b><br>PI, TESS Cycle 7 Guest Investigator, \$70k                                       | 2024–2025 |
| <b>Cool Stars Meeting 22: Support for Hybrid and Early Career Researchers</b><br>PI, NASA TWSC, \$50k                                                                                                | 2024–2025 |

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| <b>Cool Stars Meeting 22: Support for the Longest Running Independently Organized Conference in Astronomy</b><br>PI, NSF AST Conference Grant, \$50k                                          | 2024–2025 |
| <b>Cool Stars Meeting 22: Support for Early Career Researchers</b><br>PI, NAASC Support for Conferences and Workshops program, \$14k                                                          | 2024–2025 |
| <b>Machine Learning Classification for Ultracool Dwarfs</b><br>PI, NSF ACCESS Explore allocation PHY240253, 200,000.0 ACCESS Credits                                                          | 2025–2026 |
| <b>Characterizing the Lowest-mass Planet Hosts and Investigating the Potential Link between Stellar Surface Gravity and Planet Occurrence</b><br>PI, NASA XRP, \$618k                         | 2024–2026 |
| <b>A Magnitude Limited Sample of M dwarfs to Study the Super-Earth Rate across the Fully Convective Boundary</b><br>PI, TESS Cycle 5 Guest Investigator, \$70k                                | 2022–2023 |
| <b>Recalibrating Fundamental Parameters for Low-Mass Stars for Current and Future Planet Hunting Surveys</b><br>PI, XSEDE Research allocation PHY220048, 8,314,035 SUs (~\$100k)              | 2022–2023 |
| <b>Recalibrating the Ultracool Dwarfs in the Transiting Exoplanet Survey Satellite Input Catalog</b><br>PI, XSEDE Startup allocation PHY200052, 100,000 SUs (~\$1k)                           | 2022–2023 |
| <b>Infrared Gold: A Student-Centered Program to Extract, Analyze, and Disseminate 20 Years of IRTF/SpeX Point-Source Spectroscopy</b><br>Co-I, NASA ADAP (PI: Adam Burgasser), \$666,511      | 2022–2025 |
| <b>Simulating Ultracool Dwarf Populations in LSST DP0 and in the Main LSST Survey</b><br>Co-I, LSST Kickstarter Grant (PI: Adam Burgasser), \$19k                                             | 2022      |
| <b>A Search for Distant Ultracool Dwarfs in Hubble Space Telescope Spectral Surveys</b><br>PI, XSEDE Startup allocation PHY200052, 50,000 SUs (~\$1k)                                         | 2020–2021 |
| <b>Planetary Collisions around Low-Mass Stars: Constraining the Timescale for Collisions and Testing the Origin of the Kepler Dichotomy</b><br>PI, NASA Sagan Postdoctoral Fellowship, \$349k | 2019–2022 |
| <b>Spectroscopic Analysis of Ultracool Dwarfs</b><br>Co-I, SDSS FAST (PI: Adam Burgasser), \$60K                                                                                              | 2016–2017 |
| <b>Low-mass Field Stars with Infrared Excesses: Possible Signatures of Planetary Collisions</b><br>Co-I, NASA ADAP (PI: Andrew West; Admin PI: Phil Muirhead), \$125K                         | 2016–2017 |

**TELESCOPE  
TIME AWARDED  
(ORDERED BY  
APERTURE)**

|                                                                                                                              |           |
|------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Keck I &amp; II 10-meters</b>                                                                                             |           |
| PI: “Abundances and Kinematics of Ultracool Dwarf Planet Host Twin Stars”<br>• 7 nights (NIRSPEC)                            | 2022–2025 |
| Co-I: “Newly Discovered Planetary Mass Objects in the Solar Neighborhood ”<br>• 4 nights (NIRES)                             | 2023–2025 |
| Co-I: “Unresolved Binaries in the Core of the Orion Nebula Cluster”<br>• 3 night (NIRSPAO)                                   | 2023–2025 |
| Co-I: “Very High Velocity Stars and Brown Dwarfs Uncovered by Backyard Worlds ”<br>• 2 nights (NIRES)                        | 2024–2025 |
| PI: “Precise Abundances of Ultracool Dwarfs using FGK Wide Binaries”<br>• 2 nights (NIRSPEC)                                 | 2024      |
| Co-I: “Keck/NIRSPEC Cadence Program: NIRSPEC Observations of Ultrashort Period Ultracool Binaries”<br>• 0.8 nights (NIRSPEC) | 2024      |

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|------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Co-I: “Resolving Elemental Abundances of a Remarkable Hierarchical Exoplanet-host System Observed by a JWST ERS Program”                 | 2024      |
| • 0.5 nights (NIRSPAO)                                                                                                                   |           |
| Co-I: “Detailed Chemical Composition and Binarity of the Hyper Metal Poor Dwarf ” Star SDSS J0815 + 4729”                                | 2023      |
| • 0.5 nights (HIRES)                                                                                                                     |           |
| Co-I: “The Old and the Quick: A Search for Halo Brown Dwarfs with Backyard Worlds”                                                       | 2022–2023 |
| • 2 nights (NIREs)                                                                                                                       |           |
| Co-I “Galactic Archaeology with Ultracool Dwarfs: Kinematic Structure Among L Dwarfs”                                                    | 2021–2022 |
| • 3.5 nights (NIRSPEC)                                                                                                                   |           |
| PI: “Resolving Tertiary Components of Wide, Very Low-mass Binaries with AO”                                                              | 2020–2021 |
| • 3 nights (NIRC2)                                                                                                                       |           |
| Co-I: “Searching for Our Coldest Young Neighbors with Backyard Worlds”                                                                   | 2018–2021 |
| • 10 nights (NIREs)                                                                                                                      |           |
| Co-I “Testing Pre-Main Sequence Evolutionary Models in the First 10 Myr”                                                                 | 2020–2021 |
| • 2.5 nights (OSIRIS)                                                                                                                    |           |
| Co-I “Completing the Kinematic Census of Local L and T Dwarfs”                                                                           | 2019–2020 |
| • 5.5 nights (NIRSPEC)                                                                                                                   |           |
| Co-I: “Secondary Radial Velocities in Short-Period Star/Brown Dwarf Spectral Binaries”                                                   | 2019–2020 |
| • 0.5 nights (NIRSPEC)                                                                                                                   |           |
| Co-I: “Dynamics of the Orion Nebula Cluster: Mass-Dependent Kinematics”                                                                  | 2019–2020 |
| • 6 nights (NIRSPAO)                                                                                                                     |           |
| PI: “Characterizing Low-mass Binaries and Searching for Hierarchical Triples: NIR Spectra of Low-mass, Wide, Common Proper Motion Pairs” | 2019      |
| • 1 night (NIREs)                                                                                                                        |           |
| <b>CHARA Array</b>                                                                                                                       |           |
| PI: “Investigating Radius Inflation in the Lowest Mass Planet Hosts: CHARA Observations of Teegarden’s Star”                             | 2020      |
| • 4 nights (CLASSIC)                                                                                                                     |           |
| <b>Gemini North &amp; South 8-meters</b>                                                                                                 |           |
| PI: “Abundances and Kinematics of Ultracool Dwarf Planet Host Twin Stars”                                                                | 2022–2025 |
| • 13.5 hours (IGRINS/IGRINS2)                                                                                                            |           |
| PI: “High-resolution Near-infrared Observations of a Planetary-mass Binary”                                                              | 2023–2024 |
| • 6.8 hours (IGRINS)                                                                                                                     |           |
| Co-I: “The BASS-Ultracool Search for Isolated Giant Exoplanet Analogs”                                                                   | 2018–2019 |
| • 22 hours (GNIRS Spectrograph)                                                                                                          |           |
| Co-I: “Confirming a new L/T transition planetary-mass object in AB Doradus”                                                              | 2018      |
| • Fast turnaround single object observation (GNIRS Spectrograph)                                                                         |           |
| <b>James Webb Space Telescope 6.5-meter</b>                                                                                              |           |
| Co-I: Cycle 3: “How big can you make a planet? Spectroscopic characterization of HD 206893B”                                             | 2024      |
| • 9.9 hours (NIRSpec)                                                                                                                    |           |
| Co-I: Cycle 3: “Arcana of the Ancients: A Spectral Metallicity Survey of the Lowest-Mass Stars and Brown Dwarfs”                         | 2024      |
| • 83.2 hours (NIRSpec and MIRI)                                                                                                          |           |
| Co-I: Cycle 2: “Establishing the Formation of AF Lep b with NIRCams: The Lowest-Mass Imaged Exoplanet with a Dynamical Mass”             | 2023      |
| • 6.5 hours (NIRCams)                                                                                                                    |           |
| Co-I: Cycle 1: “Direct Imaging Spectroscopy of Two Jovian Exoplanets: Characterization of the TYC 8998-760-1 Multi-Planetary System”     | 2021      |
| • 5.2 hours (NIRSpec and MIRI)                                                                                                           |           |

**LDT 4.3-meter**

- PI: “Pre-main Sequence or Field Stars?: Searching for Traces of Youth in Low-mass Stars with Extreme Mid-infrared Excesses” 2016
- 2 nights (DeVeny Optical Spectrograph)

**CFHT 3.6-meter**

- Co-I: “Precision NIR RVs for WISE J1624-3212: A Nearby, Potentially Unresolved Low-mass Binary” 2021
- 2.4 hours (SPIRou NIR Spectrograph)

**IRTF 3.2-meter**

- PI: “Characterizing the Ultracool *TESS* Targets: Investigating the Role of Gravity in Planet Hosts” 2020–2025
- 421 hours (SpeX NIR Spectrograph)
- Co-I: “Characterizing Cool Hosts of Candidate Transiting Exoplanets with IRTF/SpeX” 2024–2025
- 6 nights (SpeX NIR Spectrograph)
- Co-I: “Characterizing the Ultracool Dwarf Benchmark Wide Binaries” 2024–2025
- 1 night (SpeX NIR Spectrograph)
- Co-I: “Homogeneous stellar characterization for M dwarfs with confirmed giant planets” 2023
- 1 night (SpeX NIR Spectrograph)
- Co-I: “Radial Velocity Monitoring of an Exceptionally Short-Period Very Low Mass Binary” 2022–2023
- 6 nights (iSHELL Spectrograph)
- Co-I: “Searching for Hierarchical Triples in Wide, Common Proper Motion, Very Low-Mass Binaries” 2018–2020
- 4.5 nights (SpeX NIR Spectrograph)
- Co-I: “Training the Cannon: Calibrating APOGEE Observations of Ultracool Dwarfs” 2018–2020
- 6 nights (iSHELL Spectrograph)
- Co-I: “LaTE-MoVeRS: New Nearby Very Low-Mass Stars and Brown Dwarfs Verified by Proper Motion from SDSS+2MASS+ *WISE*” 2017–2019
- 4.5 nights (SpeX NIR Spectrograph + MORIS)

**Shane 3-meter**

- PI: “Optimizing Target Selection of Direct Imaging Planet Campaigns using Accelerating Stars” 2022–2024
- 9 nights (ShARCS AO NIR Imager)
- Co-I: “Optical Spectroscopy of New and Benchmark M & L Dwarfs from Gaia and Backyard Worlds, and Potential Exoplanet Hosts from TESS and SPECULOOS” 2022–2024
- 31 nights (Kast Optical Spectrograph)
- PI: “AO Observations of Overluminous members of Wide, Low-mass Binaries: Searching for Hierarchical Triples” 2019–2021
- 17 nights (ShARCS AO NIR Imager)
- Co-I: “Optical Spectroscopy of LaTE-MoVeRS M and L Dwarfs” 2017–2022
- 63 nights (Kast Optical Spectrograph)

**SDSS 2.5-meter**

- Co-I: “APOGEE-2 Survey of the Lowest-mass Stars and Brown Dwarfs: Composition, Chemistry and Companions” 2017–2018
- ~500 APOGEE fibers awarded for ancillary science call

**Hubble Space Telescope 2.4-meter**

- PI: Cycle 32 (& 33) - “It Takes Two Planets to Tango: Constraining the Orbit of a Planetary-Mass Binary” 2024– 2025
- 6 (+4) orbits (WFC3)
- Co-I: Cycle 30 - “Completing the stellar census of Westerlund 1” 2022
- 7 orbits (WFC3)

|  |                                                                                                                                                      |              |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|  | <b>APF 2.4-meter</b>                                                                                                                                 |              |
|  | Co-I: “Abundances of Directly Imaged Planet Host Stars”                                                                                              | 2021–2024    |
|  | • 3 nights (Levy Optical Spectrograph)                                                                                                               |              |
|  | Co-I: “Calibrations of Chemical Abundances of Ultracool Dwarfs in Wide Binary Systems with Optical High-Resolution Spectroscopy of G-Type Primaries” | 2022         |
|  | • 1 nights (Levy Optical Spectrograph)                                                                                                               |              |
|  | Co-I: “Benchmarking Chemical Abundances of Ultracool Dwarfs in Binary Systems with Optical Spectroscopy of Bright AFGK Primaries”                    | 2022         |
|  | • 2.5 nights (Levy Optical Spectrograph)                                                                                                             |              |
|  | Co-I: “Radial Velocity Monitoring of WISE J1624-3212: A Potential Low-mass Binary Hiding at 18 pc”                                                   | 2021         |
|  | • 1 nights (Levy Optical Spectrograph)                                                                                                               |              |
|  | <b>Nickel 1-meter</b>                                                                                                                                |              |
|  | PI: “UCSD Observational Astrophysics Course: Proper Motions of Asteroids”                                                                            | 2024–2025    |
|  | • 14 nights (Direct Imaging Camera)                                                                                                                  |              |
|  | <b><i>Transiting Exoplanet Survey Satellite</i> 4×0.1-meter</b>                                                                                      |              |
|  | PI: “A Magnitude Limited Sample of M dwarfs to Study the Super-Earth Rate across the Fully Convective Boundary”                                      | 2022–2025    |
|  | • GO Cycle 5: 658 short cadence targets                                                                                                              |              |
|  | • GO Cycle 7: 621 short cadence targets                                                                                                              |              |
|  | <b>ADDITIONAL OBSERVING EXPERIENCE</b>                                                                                                               |              |
|  | <b>Keck I 10-meter</b>                                                                                                                               |              |
|  | 3 nights on the optical spectrometer (LRIS).                                                                                                         | 2009–2010    |
|  | <b>Keck II 10-meter</b>                                                                                                                              |              |
|  | 4 half-nights on the high-res NIR spectrometer (NIRSPEC) with AO (NIRSPA0).                                                                          | 2018         |
|  | <b>Palomar Hale 200-inch</b>                                                                                                                         |              |
|  | 1 night on the optical spectrograph (DBSP).                                                                                                          | 2009         |
|  | <b>CTIO SMARTS 0.9-meter</b>                                                                                                                         |              |
|  | 27 nights on the optical imager.                                                                                                                     | 2014–2016    |
|  | <b>PROFESSIONAL AFFILIATIONS</b>                                                                                                                     |              |
|  | International Astronomical Union (IAU)                                                                                                               | 2025–Present |
|  | Society of Senior Ford Fellows (SSFF)                                                                                                                | 2024–Present |
|  | Society for the Advancement of Chicanos and Native Americans in Science (SACNAS)                                                                     | 2016–Present |
|  | National Society of Black Physicists (NSBP)                                                                                                          | 2011–Present |
|  | American Astronomical Society                                                                                                                        | 2009–Present |
|  | American Physical Society (APS)                                                                                                                      | 2008–Present |
|  | National Society of Hispanic Physicists (NSHP)                                                                                                       | 2008–Present |
|  | <b>OTHER WORK EXPERIENCE</b>                                                                                                                         |              |
|  | <b>Booz Allen Hamilton (BAH)</b> , San Diego, California, USA                                                                                        | 2019         |
|  | Strategic Innovation Group - Lead Data Scientist                                                                                                     |              |
|  | Co-organizer of the 2019 Kaggle Data Science Bowl                                                                                                    |              |
|  | <b>Aeronautical Radio, Incorporated (ARINC)</b> , San Diego, California, USA                                                                         | 2007–2011    |
|  | Analyst/Network Engineer                                                                                                                             |              |
|  | <b>REFERENCES</b>                                                                                                                                    |              |
|  | <b>Dr. Quinn Konopacky</b>                                                                                                                           |              |
|  | Associate Professor of Astronomy & Astrophysics                                                                                                      |              |
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[CV compiled on 2026-06-10]